Pocket Money Expenditure Analysis

A statistical study of how college students receive, spend, save and invest their pocket money

Survey of 203 college students | Python (pandas, SciPy, NumPy, Matplotlib, Seaborn)

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Data: survey collected by Vaishnavi Deshmukh (Aug-Sep 2023), used with permission

Methods: data cleaning, descriptive statistics, exploratory visualisation, Welch t-tests, one-way ANOVA, chi-square tests of independence, Mann-Whitney and Kruskal-Wallis checks, Spearman and Pearson correlation, simple and multiple linear regression with diagnostics and a sensitivity analysis.

1. Executive summary

Question. What drives how much pocket money a college student gets, and how do students use and manage it? Data: 203 survey responses (193 with a usable monthly pocket-money amount).

- **Typical amount.** The median student gets **Rs 500** a month (mean Rs 1091; the mean is pulled up by a few large values, skewness 3.2). Half of students receive between Rs 200 and Rs 1500.
- **Where you live matters most.** Living away from parents is by far the strongest factor (one-way ANOVA, $F(3, 189) = 9.60$, $p < 0.001$). Students in a rented room or hostel receive roughly 2.6 to 3.3 times the pocket money of students living with their parents, after controlling for other factors. This is the only effect that is robust to every check performed.
- **Other differences are weaker.** Males, urban students and higher-income families receive more in some tests (p between 0.01 and 0.05), but these mostly disappear when 13 very large amounts are set aside, and with twelve tests run a p-value just under 0.05 can be a chance result.
- **Saving and investing.** About 27% of students invest some savings (mostly 'Other' options, then the stock market). Students who receive more pocket money save a smaller share of it (Spearman rho = -0.20, p = 0.004).
- **Model fit.** The multiple regression explains 21% of the variation in log pocket money (adjusted $R^2 = 0.18$); family income alone explains only 2.2%.
- **Spending pattern.** 87% spend on academic items (photocopies, projects, college events), against 27% on personal items such as food, films and trips (multiple answers allowed).

2. Data and cleaning

The data come from a bilingual (English/Marathi) online questionnaire answered by students in August-September 2023. It records demographics, family background, the amount of pocket money received, how it is spent, saved and invested, and payment habits. The file was not analysed as received; the following steps were applied (all reproducible in `src/01_clean_data.py`).

- **Privacy.** E-mail addresses, exact timestamps and free-text suggestions were removed before any analysis or publication. Department and the parents' occupations were also left out of the published data because they could identify individuals; no analysis uses them.
- **Headers.** The questionnaire has duplicate column names ("Which way?" appears three times), so columns were renamed by position and Marathi text was stripped from multi-select answers.
- **Duplicates.** Two rows have identical answers but were submitted on different days, so they were kept as separate respondents. No repeated timestamps were found.
- **Pocket money.** 14 answers given as ranges (e.g. 1000-2000) were converted to the midpoint; 10 answers (6 blank, 4 such as "only when needed") could not be quantified and were excluded from pocket-money analyses, leaving n = 193.
- **Share spent / saved.** Banded answers (e.g. "30% to 40%") were converted to band midpoints. The extreme options had been typed as 0 and 1; 1 was treated as 100%, which cross-tabulation supports. Spent + saved is within 10 points of 100% for 77% of respondents, so the share saved is used as the main savings measure.
- **Income, family size.** Income bands were coded to midpoints (Rs 20k, 40k, 60k, 85k, 110k per month); "More than 6" family members was coded 7. The two sparsest income bands (70-100k, above 100k) were merged as "70k+" for group comparisons.
- **Outliers.** 13 respondents lie above the IQR upper fence (Rs 3450). They are real answers, not errors, so they were kept; the analysis uses ln(pocket money) and a sensitivity check re-fits the model without them.

- **Constant column.** Every respondent answered "Yes" to "Do you take pocket money from parents?", so that variable carries no information and was dropped.

Assumption to note: the questionnaire does not state the time period of the pocket-money amount; it is treated as a monthly amount (the follow-up question asks whether the respondent needs the same amount "every month"). A few very small values (Rs 10-50) may refer to a different period and cannot be verified.

3. Descriptive statistics

3.1 Who responded

Respondents are 56% male, 73% from rural areas and 75% from the science stream. Most are 17-19 years old, 45% live with their parents and 56% come from families earning Rs 10,000-30,000 a month. Because the sample is concentrated in a few groups, comparisons involving small groups (living with relatives, n = 12; arts stream, n = 13; income above Rs 70k, n = 13) have low power.

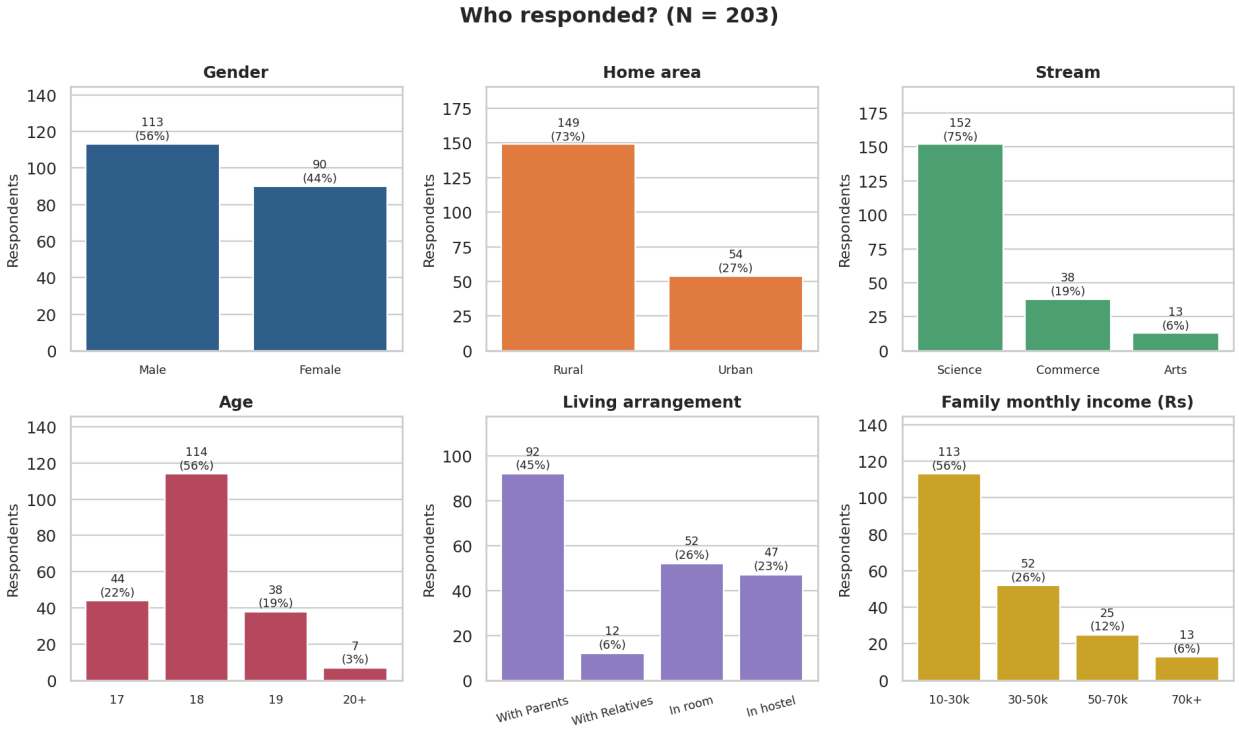


Figure 1. Profile of the 203 respondents.

3.2 Numeric variables

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max	Skew
Pocket money received (Rs/month)	193	1091	1541	10.0	200	500	1500	10000	3.23
ln(pocket money)	193	6.24	1.31	2.30	5.30	6.21	7.31	9.21	-0.24
Share saved (%)	203	40.3	25.6	0.00	15.0	35.0	55.0	95.0	0.37
Share spent (%)	203	53.1	28.3	0.00	35.0	55.0	75.0	100	0.17
Family income (Rs '000)	203	34.5	20.0	20.0	20.0	20.0	40.0	110	1.46
Family size	203	4.92	1.22	2.00	4.00	5.00	6.00	7.00	0.36
Earners in family	203	1.51	0.79	1.00	1.00	1.00	2.00	5.00	1.71

Pocket money is strongly right-skewed (skewness 3.23; coefficient of variation 141%) and fails a normality test (Shapiro-Wilk $p < 0.001$). After a log transformation the skewness falls to -0.24 and the shape is close to

normal (Figure 2), although the Shapiro-Wilk test still detects small departures (p = 0.001). All parametric tests below therefore use ln(pocket money), with non-parametric tests reported as a robustness check. The geometric mean (Rs 515) is a better 'typical' value than the arithmetic mean.

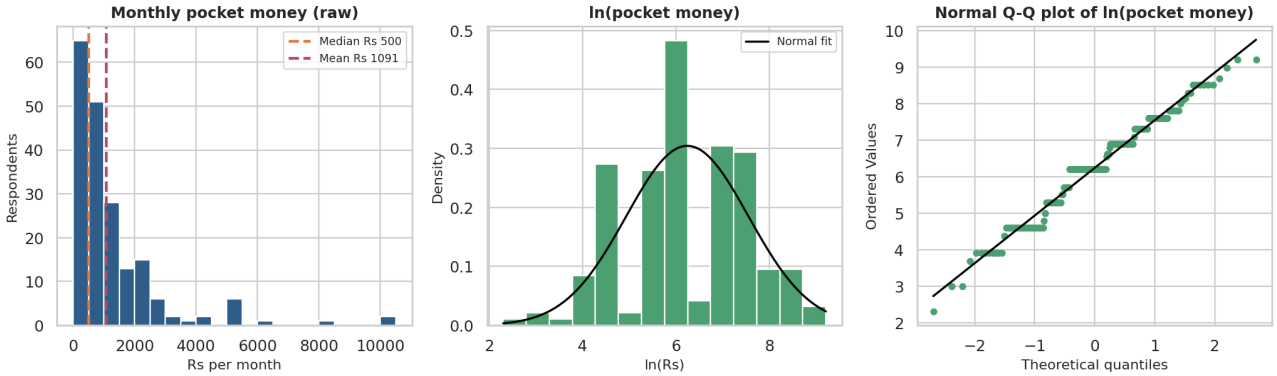


Figure 2. Pocket money before and after log transformation, with Q-Q plot.

3.3 Behaviour and payment habits

Pocket money is mostly used for academic items (87%); 27% report personal spending and 8% lend money to friends. PhonePe is the most-used payment app (64%), and 6% reported no app. On average respondents save about 40% of their pocket money; 8% save nothing. 27% invest, and 71% say they need the same amount every month.

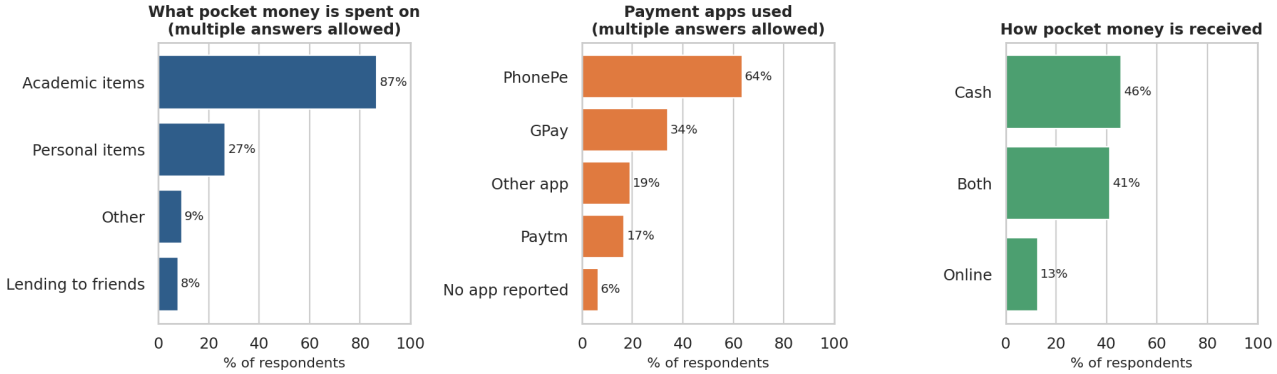


Figure 3. Spending categories, payment apps and mode of receiving pocket money.

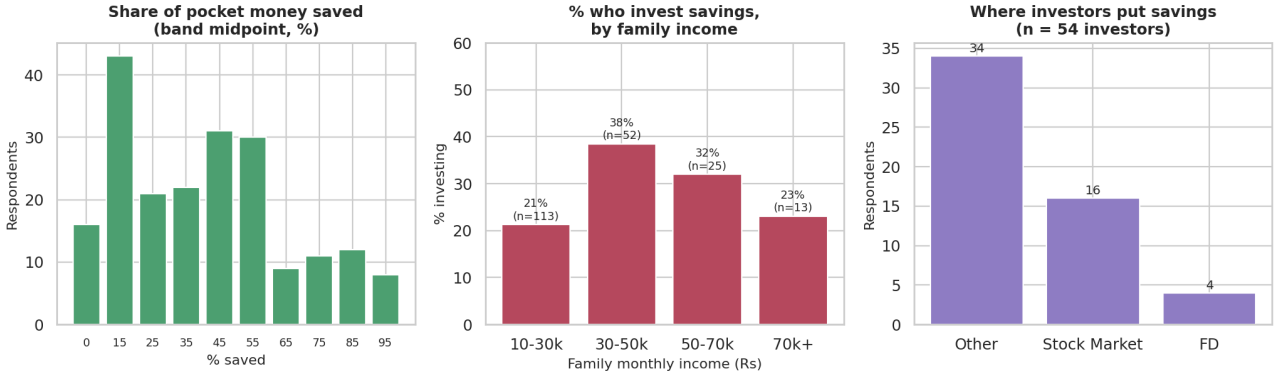


Figure 4. Savings share, investment rate by family income, and investment choices.

4. Exploratory analysis: pocket money by group

Figure 5 and the table show how pocket money differs across groups (log scale). The clearest visual gap is between students living with parents (median Rs 250) and those living in a room or hostel (median Rs 500-1,000). Differences by gender, home area and family income are smaller and overlap heavily, apart from the small top income band.

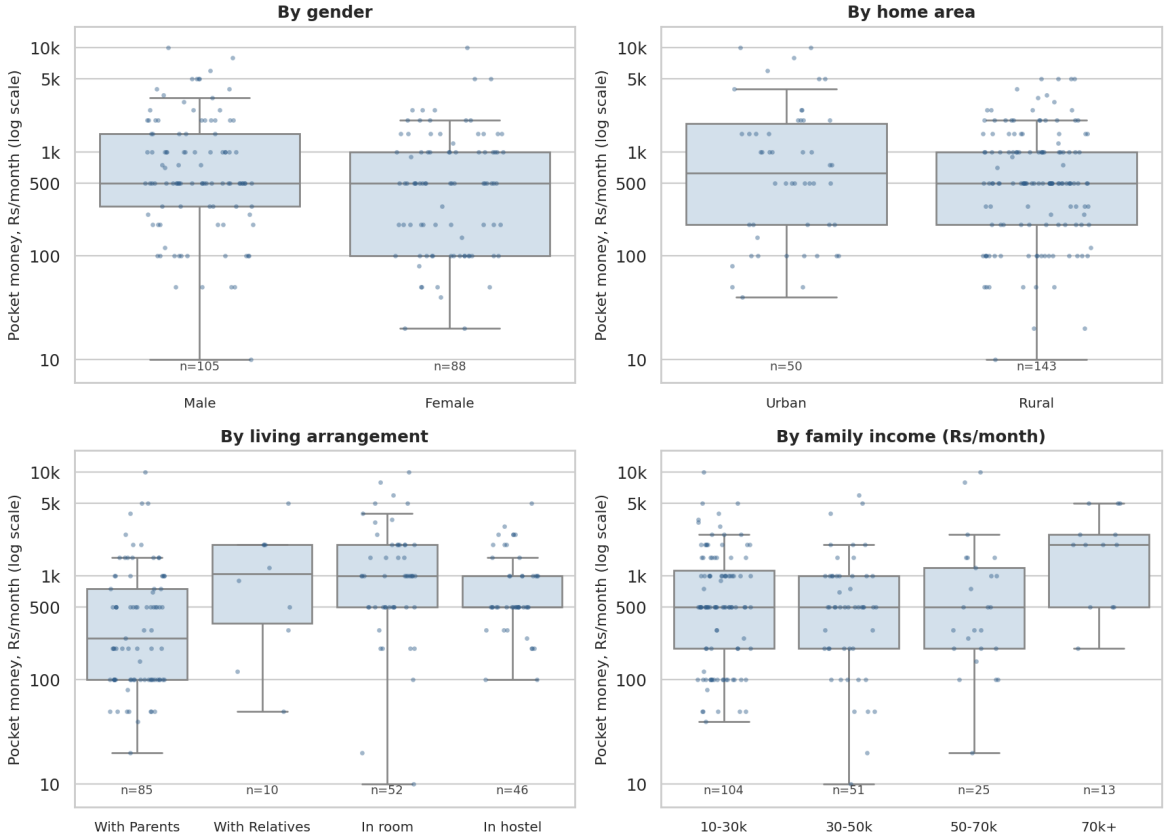


Figure 5. Pocket money by gender, home area, living arrangement and family income (log scale).

Group	n	Median (Rs)	Mean (Rs)	Geo-mean (Rs)
gender: Female	88	500	891	415
gender: Male	105	500	1258	617
origin: Rural	143	500	908	485
origin: Urban	50	625	1611	611
living: In hostel	46	500	999	681
living: In room	52	1000	1664	888
living: With Parents	85	250	752	305
living: With Relatives	10	1050	1407	723
income_group: 10-30k	104	500	1002	501
income_group: 30-50k	51	500	861	424
income_group: 50-70k	25	500	1307	483
income_group: 70k+	13	2000	2285	1556
stream: Arts	11	200	436	217
stream: Commerce	33	500	802	421
stream: Science	149	500	1203	574

5. Hypothesis tests

All tests use a 5% significance level. Twelve tests were run, so a p-value just under 0.05 could be a chance result; conclusions therefore lean on effects that stay consistent across several checks. 'Parametric' tests were applied to ln(pocket money); the non-parametric equivalent is shown in the notes as a robustness check.

ID	Test	Comparison	p	Significant at 5%?
T1	Welch t-test	ln(pocket money): Male vs Female	0.0364	Yes
T2	Welch t-test	ln(pocket money): Urban vs Rural	0.3294	No
A1	One-way ANOVA	ln(pocket money) by living arrangement (4 groups)	<0.001	Yes
A2	One-way ANOVA	ln(pocket money) by family income band (4 groups)	0.0139	Yes
A3	One-way ANOVA	ln(pocket money) by stream (3 groups)	0.0363	Yes
M1	Mann-Whitney U	Share of pocket money saved: away from family vs with family	0.6700	No
C1	Chi-square	Invests savings x Gender	0.0870	No
C2	Chi-square	Invests savings x Home area (rural/urban)	0.3472	No
C3	Chi-square	Invests savings x Family income (3 bands)	0.0662	No
C4	Chi-square	Needs same amount monthly x Living (family/away)	0.6894	No
C5	Chi-square	Spends on personal items x Gender	0.5349	No
C6	Chi-square	Mode of receiving pocket money x Home area	0.1708	No

5.1 Welch t-tests (two groups)

- **T1 - Gender.** Hypotheses: H0 mean ln(pocket money) is equal for males and females. Males receive more: geometric-mean ratio 1.49 (95% CI 1.03 to 2.16); t(183.2) = 2.11, p = 0.036. The Mann-Whitney check is borderline (p = 0.051), so this is suggestive, not conclusive.
- **T2 - Home area.** No evidence of a difference between urban and rural students: ratio 1.26 (95% CI 0.79 to 2.02), t(73.9) = 0.98, p = 0.329. Welch's version was used because it does not assume the two groups have equal spread.

5.2 One-way ANOVA (three or more groups)

- **A1 - Living arrangement.** Strong evidence of differences: F(3, 189) = 9.60, p < 0.001. Median pocket money is Rs 250 for students living with parents, Rs 500 in a hostel and Rs 1,000 in a rented room. The Kruskal-Wallis test, which does not need bell-shaped data, was also run (p < 0.001) and agrees.
- **A2 - Family income.** F(3, 189) = 3.63, p = 0.014. The difference comes from the 70k+ group (n = 13; geometric mean Rs 1,556 against Rs 424-501 in the other bands); the three lower bands look alike. With only 13 students in the top band this is not conclusive.
- **A3 - Stream.** F(2, 190) = 3.37, p = 0.036. Arts students receive the least (geometric mean Rs 217 against Rs 574 for Science), but Arts has just 11 usable respondents. Treat as exploratory.

5.3 Mann-Whitney U test (savings share)

M1. The share saved is a banded, ordinal measure, so a rank test was used. Students living away from family save a median of 35% versus 45% for those at home, which is not a significant difference (U = 4971, p = 0.670). Despite getting more pocket money, students away from home do not save a smaller share.

5.4 Chi-square tests of independence

Chi-square tests were used for pairs of categorical variables. Every table meets the usual condition that expected counts are at least 5 (smallest expected count 6.9).

ID	Association tested	n	Chi-square	df	p
C1	Invests savings x Gender	203	2.93	1	0.087
C2	Invests savings x Home area (rural/urban)	203	0.88	1	0.347
C3	Invests savings x Family income (3 bands)	203	5.43	2	0.066
C4	Needs same amount monthly x Living (family/away)	203	0.16	1	0.689
C5	Spends on personal items x Gender	203	0.39	1	0.535
C6	Mode of receiving pocket money x Home area	203	3.53	2	0.171

None of the six associations is statistically significant. The closest is investing by family income (C3, p = 0.066): the share of students who invest is about 21% in the lowest income band, 38% in the middle band and 29% in the highest, which is not a clean upward trend, and the sample is too small to confirm any pattern. Gender, home area and living arrangement are not associated with investing, needing a fixed monthly amount, personal spending, or the mode (cash/online) in which pocket money is received.

6. Correlation analysis

Spearman rank correlations were used because pocket money is skewed and several variables are ordinal. Figure 6 shows the matrix; all pairs are saved in `results/correlation_pairs_ranked.csv`.

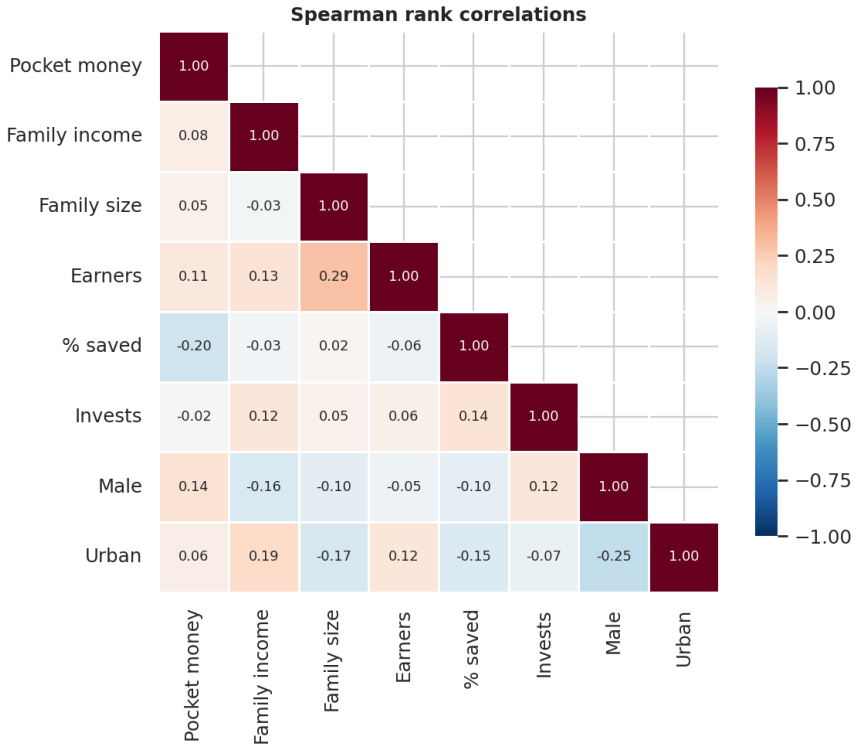


Figure 6. Spearman correlation matrix (lower triangle).

- **Pocket money vs % saved: rho = -0.20 (p = 0.004).** A weak negative link: students who receive more save a smaller share of it. This is the clearest correlation in the data.
- **Pocket money vs family income:** Spearman rho = 0.08 (p = 0.295); Pearson r on ln(pocket money) = 0.15 (p = 0.038). Family income is only very weakly related to pocket money, a recurring theme in this study.
- **Family income vs earners:** rho = 0.13 (p = 0.058), borderline; income is unrelated to family size (rho = -0.03).
- **Correlation is not causation.** These are associations in a cross-sectional sample.

7. Regression analysis

The response variable is ln(pocket money); coefficients are therefore interpreted as percentage changes: a coefficient b means a change of (exp(b) - 1) x 100% in pocket money. Family income is expressed per Rs 10,000 of monthly income.

7.1 Simple linear regression: ln(pocket money) on family income

Each additional Rs 10,000 of family monthly income is associated with a 10.1% higher pocket money (95% CI 0.5% to 20.6%; p = 0.038). However R^2 is only 0.022: family income explains about 2.2% of the variation, so it is a poor predictor on its own.

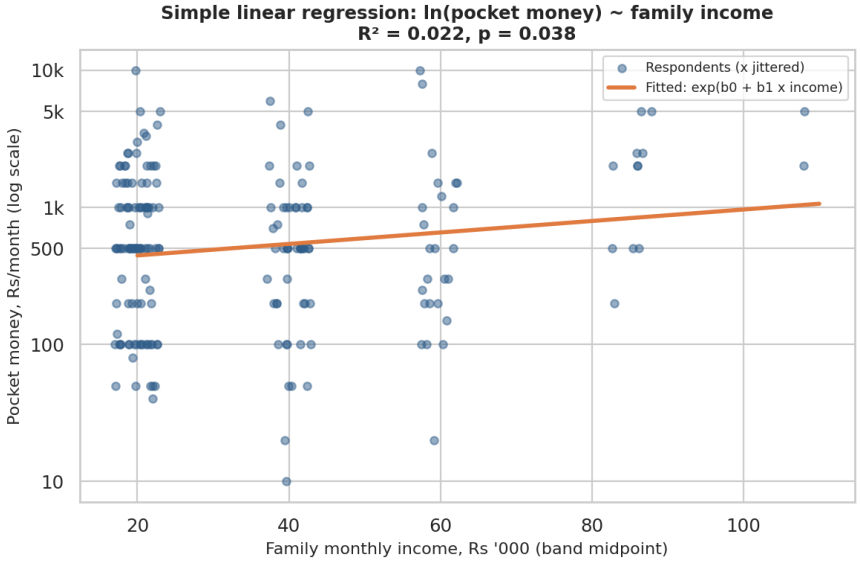


Figure 7. Fitted simple regression on the original Rs scale (log axis).

7.2 Multiple linear regression

Predictors: family income, family size, number of earners, gender, home area and living arrangement (reference: lives with parents). n = 193.

Term	Coef (ln scale)	SE	p	p (robust HC3)	% change in pocket money (95% CI)
Intercept	4.4766	0.4468	<0.001	<0.001	-
Family income (Rs '000)	0.0121	0.0044	0.007	0.014	13% (3 to 23)
Family size	0.0532	0.0805	0.509	0.560	5% (-10 to 24)
Earners	0.1091	0.1302	0.403	0.473	12% (-14 to 44)
Male	0.3804	0.1872	0.044	0.051	46% (1 to 112)
Urban	0.4945	0.2179	0.024	0.047	64% (7 to 152)
Lives in hostel*	0.9492	0.2218	<0.001	<0.001	158% (67 to 300)
Lives in a room*	1.1840	0.2250	<0.001	<0.001	227% (110 to 409)
Lives with relatives*	0.6956	0.4017	0.085	0.140	100% (-9 to 343)

Fit: R^2 = 0.214, adjusted R^2 = 0.180, F = 6.28 (p < 0.001), residual standard error = 1.19 (log units). Living in a room or hostel has by far the largest effect: holding other variables fixed, pocket money is about 158% higher (hostel) and 227% higher (room) than for students living with parents. Family income (about +13% per Rs 10,000), male gender (+46%) and urban home (+64%) are significant at 5%, whereas family size and number of earners are not. Living with relatives is not significant, but only 10 respondents are in that group, so the confidence interval is very wide.

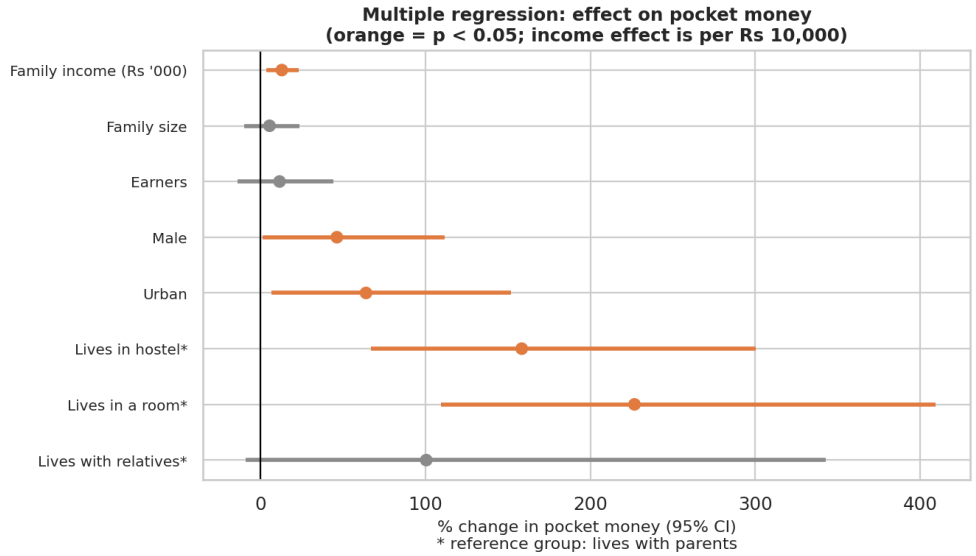


Figure 8. Estimated effects with 95% confidence intervals.

7.3 Model diagnostics

- **Multicollinearity:** all variance inflation factors are below 1.36 (rule of thumb: concern above 5-10).
- **Constant variance:** Breusch-Pagan test p = 0.759; no evidence of heteroscedasticity. Robust HC3 standard errors give the same conclusions (except that the male effect is borderline, p = 0.051).
- **Normality of residuals:** the Q-Q plot is close to a straight line, with a few very small pocket-money values in the lower tail; Shapiro-Wilk detects a mild departure (p = 0.024). With n = 193 and consistent robust results this is not a serious concern.
- **Influence:** 11 observations exceed the 4/n guideline for Cook's distance, but the largest value is 0.10, far below 1, so no single respondent drives the results.

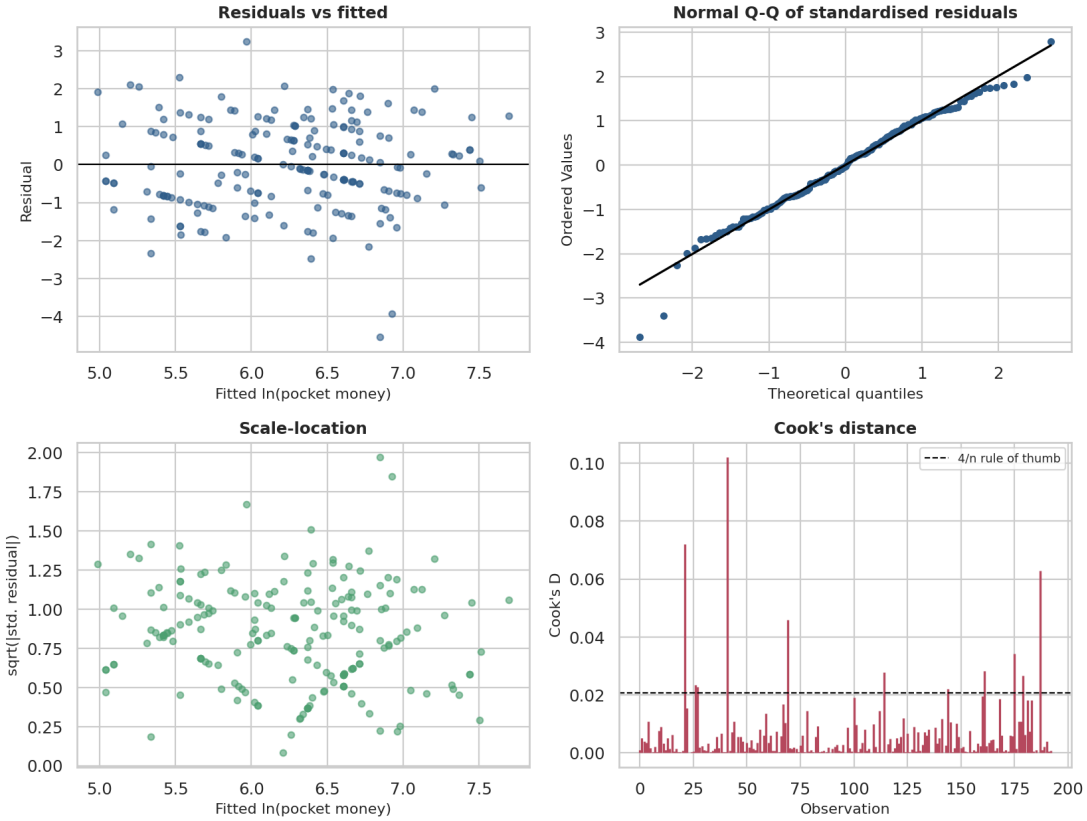


Figure 9. Diagnostic plots for the multiple regression.

8. Sensitivity analysis: are the results robust?

The model was re-fitted after removing the 13 respondents above the IQR fence (pocket money above Rs 3450), leaving n = 180.

Term	Full model: % change (p)	Without large amounts: % change (p)
Family income (Rs '000)	13% (0.007)	8% (0.105)
Family size	5% (0.509)	1% (0.942)
Earners	12% (0.403)	12% (0.379)
Male	46% (0.044)	23% (0.254)
Urban	64% (0.024)	22% (0.362)
Lives in hostel*	158% (<0.001)	161% (<0.001)
Lives in a room*	227% (<0.001)	173% (<0.001)
Lives with relatives*	100% (0.085)	104% (0.070)

The effect of living in a room or hostel is almost unchanged (about +160% to +173%) and stays highly significant. By contrast, the effects of family income, gender and home area shrink and become non-significant (R^2 falls to 0.18). These three effects therefore depend on a small number of respondents who receive large amounts and should be described as tentative. This agrees with the ANOVA, where the income effect was driven by the 13 respondents in the top income band.

9. Conclusions

- **Living arrangement is the dominant driver of pocket money.** Students living in a room or hostel receive roughly 2.6-3.3 times as much as those living with parents, regardless of income, gender or home area. A plausible explanation is that they must cover food, rent and travel themselves, but the survey cannot prove this.
- **Family income, gender and home area matter much less than expected.** They show small or borderline effects that mostly disappear when the large amounts are removed.
- **Saving and investing are largely unrelated to demographics.** No significant association was found between investing and gender, home area or living arrangement; students with more pocket money tend to save a smaller share.
- **Practical implication.** A college or financial-literacy programme looking for students who need money-management support should target students living away from home, who handle the largest amounts.

10. Limitations

- **Convenience sample** of students from one institution (answered voluntarily); the results cannot be generalised to all students.
- **Self-reported, banded data.** Percent spent/saved and family income are in bands; midpoints add measurement error.
- **Unstated time period** for pocket money (assumed monthly) and a few implausibly small values.
- **Small subgroups** (living with relatives n = 10-12, arts n = 11-13, income above Rs 70k n = 13) limit statistical power.
- **Observational, cross-sectional data:** associations are not causal. Several predictors are related (e.g. gender and living arrangement), which regression only partly separates.
- **Multiple comparisons:** twelve tests were run, so an isolated p-value just under 0.05 may be a chance finding.

11. Reproducibility

Run `python src/01_clean_data.py`, then `python src/02_analysis.py`, then `python src/03_build_report.py`. All tables are in `results/` and all figures in `figures/`. OLS, Breusch-Pagan, HC3 errors, VIF and Cook's distance are implemented directly with NumPy/SciPy in `src/stats_utils.py`.